

Infrared Touchless Tracking Display: A Contactless Alternative to Touchscreens

Pujan Pathak
University of Texas at Arlington

Tobin Dietrich
University of Texas at Arlington

Eduardo Natera
University of Texas at Arlington

Abstract—Touchscreen interfaces have become widespread in embedded systems, particularly in public and high-traffic environments such as ATMs, self-checkout stations, and industrial control panels. However, their shared use raises concerns regarding hygiene, as frequent physical contact allows these surfaces to act as vectors for microbial transmission. This project addresses that issue by developing a touchless, infrared-based tracking system that serves as an alternative method for user interaction. The system uses infrared sensing to detect user input without requiring direct contact and is integrated with a display to replicate the functionality of a traditional touchscreen interface. A working prototype was designed and implemented to evaluate the feasibility of this approach, focusing on input detection, responsiveness, and overall usability. Initial results demonstrate that the system is capable of reliably tracking user interactions with minimal latency, supporting its viability as a contactless interface. This work suggests that infrared-based touchless displays can provide a practical and more hygienic alternative to conventional touchscreen systems without significantly reducing functionality or user intuitiveness.

Index Terms—Infrared sensing, touchless interaction, gesture tracking, embedded systems

I. INTRODUCTION

Embedded processing is central to modern interactive systems, allowing inputs from sensors to be linked to processors for real-time operation. Touchscreens have become the most common form of input in these systems because they are intuitive to use and require little to no prior experience. They also simplify interface design in embedded applications. However, their reliance on physical contact introduces concerns related to hygiene, especially in public settings where many users interact with the same surface.

Recent research has highlighted infection risks associated with shared surfaces, strengthening the need for alternative interaction methods. One quantitative risk assessment demonstrated that public touchscreens can act as “fomites,” or objects likely to carry and transmit infection between users [1]. This issue motivates the development of new interaction systems that reduce or eliminate the need for direct contact.

Advancements in sensor technology and embedded computing have enabled the development of touchless interaction methods such as infrared tracking and gesture recognition. One study presents a touchless user interface based on near-infrared-sensitive imaging that can be integrated on top of a display, supporting gesture-based interaction without physical contact [2]. In addition, surveys of gesture-controlled interfaces show that non-contact input systems are an active area

of research and highlight their potential for intuitive user interaction [3]. Although these works demonstrate the feasibility of touchless systems, they primarily focus on sensing techniques rather than the design of a complete embedded display system intended for public use.

This project addresses that gap by developing a touchless infrared-based tracking system integrated with a display. The system is designed to replicate the functionality of a traditional touchscreen while removing the need for physical contact. The main contribution of this work is the design and implementation of a real-time, embedded touchless display, demonstrating a practical approach to contactless interaction in shared environments.

II. SYSTEM MODEL

A. Overview

The system is designed as a touchless input interface that uses an infrared (IR) sensing grid to detect user interaction and map it to a visual output on a display. The overall architecture consists of three main components: an infrared sensing frame, an STM32 microcontroller, and a TFT LCD display, as shown in figure 1.

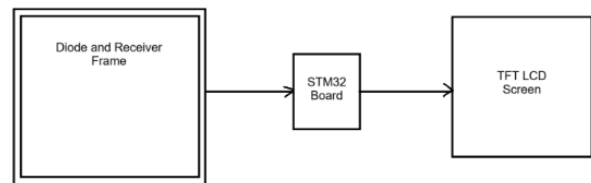


Fig. 1. Block diagram of the touchless infrared sensing and display system.

The infrared sensing frame is constructed using a 2D grid of IR emitters and IR receivers arranged along the X and Y axes. The emitters continuously transmit infrared beams across the frame, which are received by aligned phototransistors. When an object interrupts a beam, the corresponding receiver detects a loss of signal, indicating the presence of an object at that coordinate. By scanning across both axes, the system determines the position of the interaction within the grid.

The IR grid is interfaced with the STM32 microcontroller through GPIO pins, with shift registers used to reduce the number of required input pins and streamline data acquisition. The shift registers allow sequential reading of multiple

sensor states, enabling efficient sampling of the grid without exceeding hardware limitations of the microcontroller.

The STM32 microcontroller serves as the central processing unit of the system. It continuously samples the IR grid using a polled sampling loop synchronized with the emitter scan to ensure consistent timing and real-time responsiveness. The collected sensor data is processed to determine the coordinates of detected interactions. These coordinates are then mapped to corresponding positions on the display.

The visual output is handled by a TFT LCD screen connected to the microcontroller via the SPI communication protocol. SPI enables high-speed, synchronous data transfer, allowing the system to update the display in real time. Based on the processed input coordinates, the microcontroller updates the display by drawing or modifying pixels at the corresponding locations, effectively replicating the behavior of a touchscreen interface without requiring physical contact.

Overall, the system operates as a pipeline in which infrared beam “interruptions” are detected by the sensing frame, transmitted to the microcontroller for processing, and then rendered visually on the display. This architecture enables real-time, touchless interaction using relatively simple hardware and efficient embedded processing.

B. Circuit Design

The hardware implementation of the system is shown in Fig. 2.

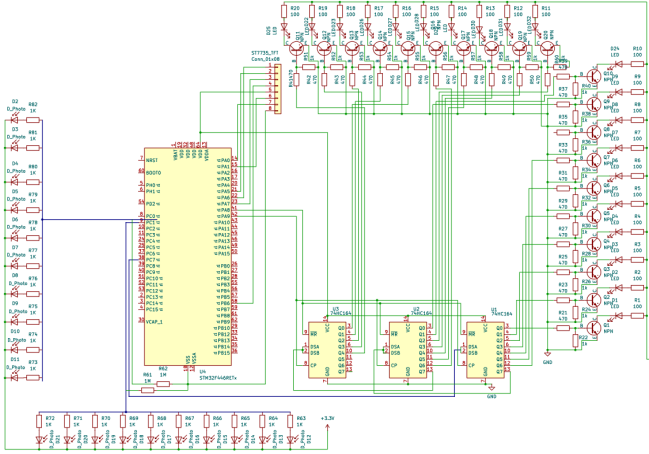


Fig. 2. Circuit of the touchless infrared sensing and display system.

The infrared sensing frame consists of twenty IR emitter–receiver pairs arranged in two perpendicular rows: ten pairs span the horizontal axis and ten span the vertical axis, forming a 10×10 sensing grid across the active area of the display. To control the twenty IR emitters without dedicating an equivalent number of GPIO pins on the STM32, the emitters are driven through a pair of cascaded 8-bit serial-in, parallel-out shift registers. The microcontroller drives the registers using only three GPIO lines: a serial data line on PC7, a clock line on PA9, and an asynchronous clear line on PA8. At the start

of each scan cycle, the firmware asserts the clear line to force every parallel output low, guaranteeing that no emitter is unintentionally active. It then shifts a single logic-high bit into the chain and clocks it into the position corresponding to the emitter under test, so that exactly one IR LED is illuminated at any moment. Because only one beam is active per sample, adjacent beams cannot interfere with one another and the receiver state can be unambiguously associated with a single grid coordinate.

On the receiver side, the ten phototransistors along each axis are wired in parallel onto a single analog line, yielding two shared receiver nodes routed to PC0 and PC1. These pins are configured as analog inputs and connected to channels IN10 and IN11 of the STM32’s onboard 12-bit ADC1 peripheral. The receiver-side multiplexing is made possible by the emitter scheme described above: since exactly one emitter is illuminated at any time, only the aligned phototransistor on the corresponding axis contributes meaningfully to the shared line, and the other nine remain in an inactive state. After each emitter is activated, the firmware allows a short settling interval, performs a single ADC conversion on the appropriate channel, and compares the result against a fixed threshold of approximately 3.0 V (a raw value of 3723 on the 0–4095 scale). A reading below the threshold indicates that the beam has been interrupted and is recorded as a blocked sensor at that grid index.

The display subsystem is connected to the STM32 over a four-wire SPI interface using the SPI1 peripheral. SCK and MOSI are mapped to PA5 and PA7 respectively, and two additional GPIO lines (PA0 and PA1) serve as the data/command select and reset signals for the ST7735 display controller. The chip-select line on PB6 is managed in software, allowing the microcontroller to share the SPI bus deterministically and issue command and data bursts as needed during pixel updates.

Overall, this design balances pin economy with sampling determinism. The shift-register chain reduces the emitter-side pin count from twenty to three, the shared ADC lines reduce the receiver-side pin count from twenty to two, and the use of an onboard ADC removes the need for additional comparator or thresholding hardware. The result is a complete sensing and display interface that fits within a small fraction of the STM32’s available I/O resources.

III. CONCLUSION

This project demonstrated the design and implementation of a touchless infrared-based tracking system as an alternative to traditional touchscreen interfaces. The system successfully showed that embedded infrared sensing can be used to detect user input and enable real-time interaction without requiring physical contact. This supports the feasibility of touchless interfaces as a practical solution for reducing hygiene concerns in shared environments.

However, the final implementation revealed limitations in input fidelity and resolution. Due to design choices in the sensing layout and system configuration, the accuracy of user input detection was lower than intended, which affected the

overall responsiveness and precision of the interface. These limitations highlight the importance of sensor density and system calibration in achieving a reliable touchless interaction experience.

Future work will focus on improving the resolution and accuracy of the system by refining the infrared sensor arrangement and increasing the density of the sensing frame. A second iteration of the design could also explore more advanced signal processing or calibration techniques to improve input tracking. With these improvements, the system has the potential to more closely match the performance and intuitiveness of traditional touchscreens while maintaining the benefits of contactless interaction.

REFERENCES

- [1] S. A. Julian et al., “Quantitative risk assessment of public touchscreens as fomites,” PubMed, 2022. [Online]. Available: <https://pubmed.ncbi.nlm.nih.gov/35333886/>
- [2] J. Kim et al., “Touchless user interface using near-infrared-sensitive imaging,” Nature Electronics, 2023. [Online]. Available: <https://www.nature.com/articles/s41928-023-00970-8>
- [3] “A survey on gesture-controlled interfaces,” International Journal of Scientific Research in Science and Technology (IJSRST), 2025. [Online]. Available: <https://ijsrst.com/index.php/home/article/view/IJSRST251222672>